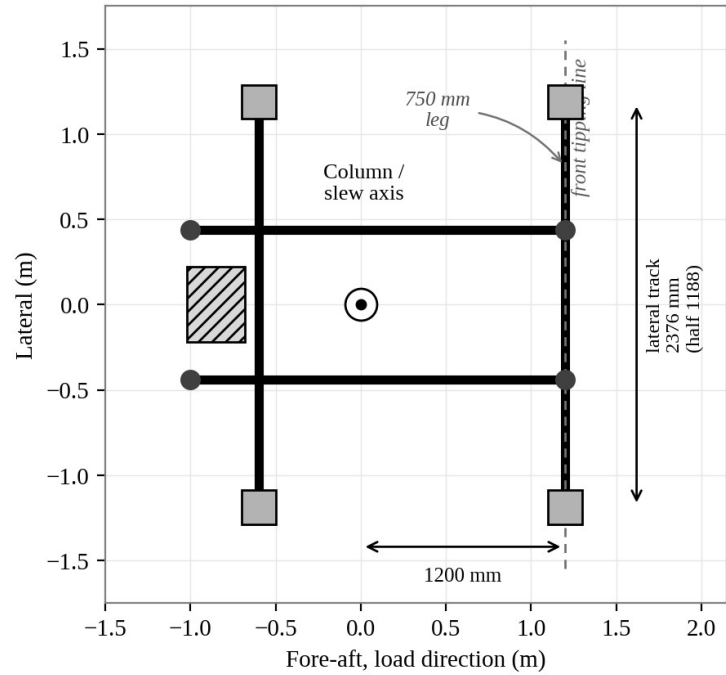


STRUCTURAL VERIFICATION CONFIGURATION COMPARISON

1 Tonne Mobile Floor Jib Crane

Wheels-only (Rev 2) against deployed outriggers (Rev A): key differences and findings

Deployed footprint, plan view: two rails plus four transverse legs (# arrangement)



Deployed footprint in the # arrangement. The legs stand square to the rails, so they widen the lateral track to 2376 mm but leave the front line at 1200 mm.

Field	Detail
Project reference	40467361
Configurations compared	Outriggers stowed (Rev 2) and deployed (Rev A, # arrangement)
Rated capacity	1000 kg at 1400 mm reach
Document revision	Comparison summary, Rev 2 against Rev A
Date	10 June 2026
Status	Issued for review
Analysis tool	MATLAB R2026a (parametric script)

Document Control

This note sets out the key differences and the findings shared between the two configuration reports for the 1 tonne mobile jib crane, project 40467361. The wheels-only Rev 2 report assesses the crane standing on its four wheels with the outrigger legs stowed. The deployed-outrigger Rev A report assesses the same crane with its four legs down in the # arrangement. Both reports are built on one parametric model and one set of inputs. This note compares them. It introduces no new analysis and supersedes neither document.

Source documents

Report	Date	Configuration	Length
Rev 2	09 Jun 2026	Wheels only, outriggers stowed	27 pp
Rev A	10 Jun 2026	Deployed outriggers, # arrangement, 90° transverse	16 pp

Basis and limitations

- **Same crane, same chart.** Both reports cover the unit on the manufacturer drawing set and check it against the same rated chart: 1000 kg at 1400 mm, 700 kg at 2320 mm and 300 kg at 3300 mm.
- **Same inputs.** Section sizes, member lengths, weld throats, bolt grade, the slewing pattern and the leg dimensions are read off the drawings. The 700 kg self-weight and the 0.30 m centre-of-gravity offset are client-supplied or estimated and carry through both reports.
- **Same targets.** Members are checked against yield to a 1.5 factor, pins and bolts to 3.0, as in both parents.
- **Scope of physics.** Static lifting only. Wind, side pull and snatch loading are outside this scope, as in both reports.
- **Status.** A comparison for review. It does not replace load testing or a competent-person examination before the crane is put to work.

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1. Executive Summary

The two reports cover the same crane and the same rated chart. They differ only in the state of the four outrigger legs. Rev 2 takes the crane on its wheels with the legs stowed. Rev A takes the same crane with the legs deployed in the # arrangement, the two fore-aft rails crossed by a front and a rear transverse beam. Rev A is a companion to Rev 2: it reworks only what the legs change and carries the rest across without alteration.

One feature of the deployed layout decides the comparison. The legs stand at 90 degrees to the rails, so they reach sideways and not forward. Putting them down widens the lateral track, which is what lets the crane slew without tipping, but it adds no reach in the luffing plane. The front tipping line stays at the front frame line, 1200 mm ahead of the column, exactly where it sits on the wheels.

Three findings follow from that. Forward stability is identical in both reports: the mid-reach point, 700 kg at 2320 mm, governs and still falls short at a factor of 1.34 against the 1.50 target. Sideways stability is the case the legs exist to answer, and it has no counterpart in the wheels-only report. With the 2376 mm track the 750 mm legs give, the crane is steady close in at 3.92 but tips at mid and far reach, at 1.05 and 1.31. The deployed leg member is comfortable at 2.11, where the wheels-only outrigger arm it replaces ran at 1.38.

Nothing above the slewing bearing changes. The legs bolt to the base frame below the bearing, so the axial load and the moment that travel up into the column come from the boom and the lifted load alone. The column stays at yield, factor 0.99, its base weld stays over target at 0.85, and the boom outer section stays marginal at 1.48, the same figures in both reports.

The conclusion is therefore the same in both states. The crane cannot be signed off to the full 1 tonne chart on a working-stress basis. Deploying the legs is a precondition for slewing and it eases the front-support member, but it does nothing for forward tipping or for the column and its base weld. The route to the structural fix, a re-rate of the chart or an upgrade of the column and weld, lives in the Rev 2 report and does not depend on the legs.

Headline comparison

Item	Result	FoS	Target	Verdict
Forward tipping, worst (700 kg @ 2320 mm)	7.69 kN·m	1.34	1.50	Fail (as wheels)
Sideways tipping, worst (700 kg @ 2320 mm)	7.77 kN·m	1.05	1.50	Fail (deployed only)
Deployed leg member (RHS 60x100x6)	130 MPa	2.11	1.50	Pass
Column, combined N+M (carried over)	277 MPa	0.99	1.50	Fail
Column-to-base weld (carried over)	235 MPa	0.85	1.50	Fail
Lateral track, provided / needed	2376 / 2784 mm	—	—	Short

2. The Two Configurations

2.1 The geometric difference

The crane base is a pair of fore-aft rails set about 876 mm apart, with a wheel at each corner. Stowed, that narrow rail track is the whole footprint, and it is the reason the crane is unsafe to slew on its wheels. Deploying the legs adds two transverse beams, one near the front wheel line and one near the rear, each reaching out to a foot on either side. The four feet, with the two rails, make the # seen in plan.

The leg dimensions are read from Base DETAIL-11: a 750 mm arm, a 230 mm drop to an 80 mm foot pad, an M16 levelling screw and a 32 mm hinge pin. Because the beams sit square to the rails, the feet widen the footprint sideways only. The front foot stays level with the front frame line, 1200 mm ahead of the column, so it adds nothing to the forward overhang.

The lateral track follows from the geometry. Each foot sits half the frame track plus one leg arm out from the centreline, so the half-track is 438 plus 750, which is 1188 mm, and the full track is 2376 mm. This single number governs sideways stability and is the make-or-break input of the deployed configuration.

Quantity	Value	Source
Leg arm (lateral reach)	750 mm	DETAIL-11
Frame track, rail to rail	876 mm	base drawing
Deployed lateral track	2376 mm $(876 + 2 \times 750)$	derived
Front foot ahead of column	1200 mm (front frame line)	base drawing
Leg member section	RHS 60 x 100 x 6	drawing

2.2 Why the difference is confined to the base

The legs attach to the base frame, below the slewing bearing. Everything above that bearing, the column, the boom, the pins and the slewing bolts, is loaded by the boom and the lifted load and is indifferent to whether the base stands on wheels or on legs. So the comparison divides cleanly. The legs change stability and the members that carry the load down to the ground, and they leave the upper structure untouched. The two reports are organised on exactly that division, which is why Rev A reworks only the base and stability and reproduces the rest from Rev 2.

3. Forward Stability

Forward tipping is a moment balance about the front foot line, 1200 mm ahead of the column. With the legs at 90 degrees, that line sits in the same place as the front wheel, so the balance is identical to the wheels-only case. The restoring moment comes from the crane weight and the counterweight acting behind the line, and the overturning moment from the lifted load ahead of it.

Overturning balance about the front foot line

$$\text{Restoring } M_R = m_{\text{crane}} \times g \times (L_{\text{fwd}} + e_{\text{cg}})$$

$$= 700 \times 9.81 \times (1.200 + 0.300) = 10.30 \text{ kN.m}$$

$$\text{Overturning } M_{OT} = m_{\text{load}} \times g \times (\text{reach} - L_{\text{fwd}}) \quad [\text{static load}]$$

Load case	M_OT (kN·m)	FoS = M_R / M_OT	Verdict
1000 kg @ 1400 mm	1.96	5.25	Pass
700 kg @ 2320 mm	7.69	1.34	Fail
300 kg @ 3300 mm	6.18	1.67	Pass

The mid-reach point governs at a factor of 1.34, below the 1.50 target. This matches the wheels-only report, as it must. Transverse legs do not move the front line, so they cannot improve forward stability. This is the first and clearest finding of the comparison: deploying the legs does nothing for forward tipping.

4. Sideways Stability During Slew

Sideways tipping is the case the legs are there for, and it has no counterpart in the wheels-only report, because the crane is not meant to slew on its wheels. With the boom slewed across the rails, the crane can tip over a side foot line set at the half-track, 1188 mm from the centreline. The restoring moment is the crane weight acting at that half-track. The overturning moment is the lifted load acting at its reach, less the half-track, since the foot line is offset toward the load by that amount.

Overturning balance about a side foot line

$$\text{Restoring } M_R = m_{\text{crane}} \times g \times t_{\text{half}} = 700 \times 9.81 \times 1.188 = 8.16 \text{ kN.m}$$

$$\text{Overturning } M_{OT} = m_{\text{load}} \times g \times (\text{reach} - t_{\text{half}}) \quad [\text{static load}]$$

$$\text{Half-track from the legs: } t_{\text{half}} = 438 + 750 = 1188 \text{ mm (track 2376 mm)}$$

Load case	M_OT (kN·m)	FoS = M_R / M_OT	Verdict
1000 kg @ 1400 mm	2.08	3.92	Pass
700 kg @ 2320 mm	7.77	1.05	Fail
300 kg @ 3300 mm	6.22	1.31	Fail

The close-in lift is steady, but the mid and far points both fall below target, at 1.05 and 1.31. The mid-reach point is again the worst, and it sets how wide the feet must stand. Solving the balance for a factor of 1.50 at that point gives a half-track of 1392 mm, a full track of 2784 mm. The legs as drawn give 2376 mm, about 400 mm short. The shortfall is in the track, not in the legs themselves. The manufacturer figure marks the deployed width as adjustable toward 2870 mm, at which the worst sideways factor lifts to about 1.6. The

deployed foot positions therefore have to be measured on the machine and set to at least 2784 mm before the crane is slewed at mid or long reach.

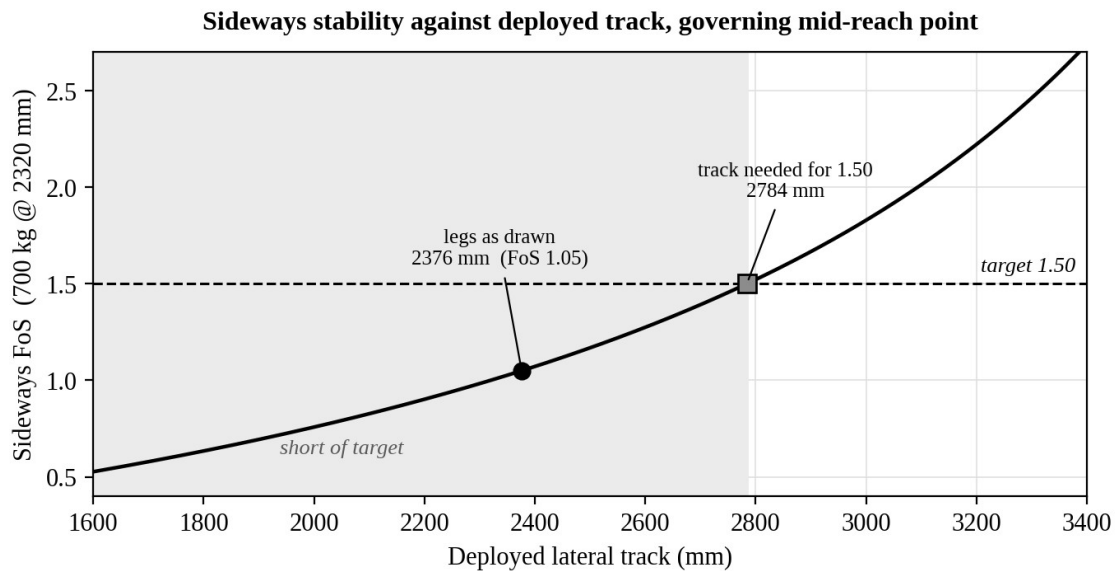


Figure 4.1 Sideways stability against the deployed track at the governing mid-reach point. The 2376 mm track from the legs sits below the 1.50 target; about 2784 mm is needed.

5. The Front Ground-Support Member

The member that carries the front reaction to the ground changes with the configuration, and it changes for the better. On the wheels, that member is the fore-aft outrigger arm, a 1200 mm cantilever that runs at 200 MPa for a factor of 1.38, a fail. Deployed, the front reaction is picked up by the leg arm instead, a shorter 750 mm cantilever in the same RHS 60 x 100 x 6 section.

Deployed leg, RHS 60 x 100 x 6, cantilever

front reaction (1000 kg @ 1400 mm) = 15.80 kN total -> 7.90 kN per leg

lever (leg arm) = 750 mm = 0.750 m

$M_{leg} = 7.90 \times 0.750 = 5.93 \text{ kN.m}$

$S_x = 45.5 \text{ cm}^3 = 45.5 \times 10^{-6} \text{ m}^3$

$\sigma = M_{leg} / S_x = 5.93 \times 10^3 / 45.5 \times 10^{-6} = 130.3 \text{ MPa}$

$FoS = f_y / \sigma = 275 / 130.3 = 2.11 \rightarrow \text{Pass}$

The leg passes at 2.11, and it is less stressed than the arm it replaces because the lever is shorter. The largest leg load comes from the heaviest lift taken in close, not from the tipping case, because member force follows the vertical load while tipping follows the lever arm. The base frame between the column and the front legs still carries that front reaction much as it does to the wheels, so the wheels-only outrigger-arm result stays representative for that member and is listed in the carry-over table in Section 6.

6. Items Carried Over Above the Slewing Bearing

The axial load and the bending moment that pass up through the bearing into the column, and on into the boom and its pins, are fixed by the boom geometry and the lifted load. None of them change when the base stands on legs instead of wheels. The wheels-only results for those members therefore apply to the deployed case without alteration. Rev A reproduces them from Rev 2 rather than reworking them, and they are collected below.

Check	FoS	Target	Verdict
Boom outer (100x140x6)	1.48	1.50	Marginal
Boom middle (80x120x6)	2.54	1.50	Pass
Boom inner (60x100x6)	3.09	1.50	Pass
Column, combined N+M	0.99	1.50	Fail
Column buckling	67.6	1.50	Pass
Base outrigger arm (fore-aft cantilever)	1.38	1.50	Fail
Pivot pin (Ø30)	2.44	3.00	Fail
Cylinder rod-end pin (Ø20 as modelled)	1.20	3.00	Fail
Hook pin (Ø20)	4.03	3.00	Pass
Slewing bolts (5 × M16, 8.8)	1.69	1.50	Pass
Weld, column base	0.85	1.50	Fail
Weld, boom bracket	3.39	1.50	Pass

The cylinder rod-end pin is listed at the conservative 20 mm diameter and fails on that basis. As Rev 2 sets out, the only hinge pin shown near the cylinder mount is the 40 mm Base Detail-4 pin, which clears the check at 4.81; the as-built diameter should be confirmed on the machine. The column at 0.99 and its base weld at 0.85 are the governing structural items in both configurations.

7. Stowed versus Deployed

The table sets the two configurations side by side. The legs change the bottom of the load path and the stability picture. They leave the top of the structure as it was.

Item	Wheels stowed	Legs deployed (# arrangement)
Front tipping line ahead of column	1200 mm	1200 mm (no forward reach)
Lateral foot track	876 mm	2376 mm
Forward tipping, worst point	FoS 1.34, Fail	FoS 1.34, Fail (unchanged)
Sideways tipping during slew	narrow track, tips	FoS 1.05, Fail (track short)
Front ground-support member	outrigger arm 200 MPa, FoS 1.38, Fail	leg arm 130 MPa, FoS 2.11, Pass
Column, combined N+M	277 MPa, FoS 0.99, Fail	unchanged
Column-to-base weld	235 MPa, FoS 0.85, Fail	unchanged
Boom outer section	186 MPa, FoS 1.48, Marginal	unchanged
Pins and slewing bolts	as Rev 2	unchanged

Read across, the pattern is consistent. The legs widen the track and ease the front-support member, so sideways tipping becomes possible to manage and the leg member passes. Forward tipping and the column, weld and boom are the same in both states, because the legs do not act on them.

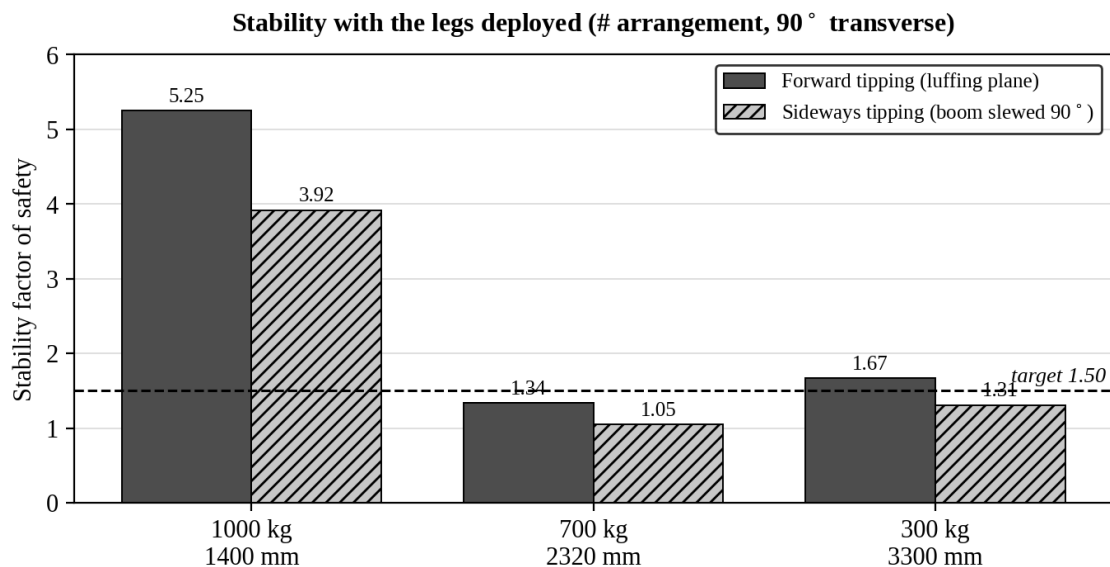


Figure 7.1 Stability in the deployed state. Forward tipping is unchanged from the wheels-only case; sideways tipping clears the target only close in.

8. Conclusions and Findings

The comparison reduces to one idea and its consequences. The four legs are transverse, square to the rails, so deploying them widens the crane sideways but not forward. From that, the two reports differ in a narrow and predictable way.

- **What the legs do.** They make slewing possible, which the crane cannot do safely on its wheels, and they ease the front ground-support member, which moves from the outrigger arm at FoS 1.38 to the shorter leg arm at FoS 2.11.
- **What the legs do not do.** They leave forward tipping exactly as it was, failing at FoS 1.34 at mid-reach, and they leave the whole structure above the slewing bearing untouched, so the column stays at 0.99 and its base weld at 0.85 in both configurations.
- **The deployed-only item.** Sideways stability is short until the track is widened. The 2376 mm track clears the target only close in; about 2784 mm is needed at mid-reach. The deployed width must be measured on the machine and, if the feet adjust, set to at least 2784 mm before any slew at mid or long reach.
- **The shared bottom line.** Neither configuration certifies the crane to the full 1 tonne chart on a working-stress basis. The structural fix, a re-rate of the chart to roughly 400 to 450 kg at mid and far reach, or an upgrade of the column section and base weld, comes from the Rev 2 report and is the same regardless of the legs.

Finding

Putting the legs down is required before any slew, and it makes the sideways case manageable once the track is set wide enough. On its own it does not make the rated chart safe. Forward tipping and the column and weld limits are identical in both states and must be resolved through the Rev 2 re-rate or upgrade. Confirm the deployed track, the leg angle, and the cylinder rod-end pin on the machine.

Standards referenced. EN 13001-3-1 (crane structural design), EN 1993-1-1 (steel structures, general), EN 1993-1-8 (joints), cross-referenced with AS 1418.1 and SS 559. Source documents: Rev 2 (wheels only) and Rev A (deployed outriggers), project 40467361, both issued for review.